



METOCEAN PROFILE FOR OGC API - EDR

STANDARD
Profile

DRAFT

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ABSTRACT

This document defines the MetOcean Profile for OGC API — EDR, an OGC API — Environmental Data Retrieval (EDR) Part 3 Service Profile. It specifies normative requirements and conformance tests for server implementations conforming to this profile.



KEYWORDS

The following are keywords to be used by search engines and document catalogues.

metocean, ogc, api, edr, weather, meteorology, high value datasets



PREFACE

This document was prepared by Norwegian Meteorological Institute, Meteorological Service of Canada.



SECURITY CONSIDERATIONS

No security considerations have been made for this document.



SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- Norwegian Meteorological Institute
- Meteorological Service of Canada



SUBMITTERS

All questions regarding this document should be directed to the editor or the submitters:

Table – Submitters

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SCOPE



SCOPE

This standard defines the MetOcean Profile for OGC API — EDR. It specifies requirements and conformance tests for implementations of this profile.



2

CONFORMANCE

Conformance with this standard shall be checked using the Abstract Test Suite in Annex A.

The following conformance classes are defined:

- <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/core>
- <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/insitu-observations>
- <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/nwp>
- <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/data-query-response-format>



3

NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Mark Burgoyne, David Blodgett, Charles Heazel, Chris Little, Open Geospatial Consortium: OGC 19-086r6, *OGC API – Environmental Data Retrieval Standard*. Open Geospatial Consortium (2023). <http://www.opengis.net/doc/IS/ogcapi-edr-1/1.1.0>.

OGC API – EDR Part 3: Service Profiles (draft)



4

TERMS AND DEFINITIONS

4

TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

This document also uses terms defined in OGC API — EDR Part 1: Core.



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REQUIREMENTS

REQUIREMENTS CLASS 1

IDENTIFIER	<code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core</code>
CONFORMANCE CLASS	Conformance class A.1: <code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/core</code>
TARGET-TYPE	MetOcean Profile for OGC API — EDR Profile Standard
NORMATIVE STATEMENTS	Requirement 1: <code>/req/core/core-openapi</code> Requirement 2: <code>/req/core/core-collection-identifier</code> Requirement 3: <code>/req/core/core-collection-title</code> Requirement 4: <code>/req/core/core-collection-license</code> Requirement 5: <code>/req/core/core-collection-spatial-extent</code> Requirement 6: <code>/req/core/core-collection-temporal-extent</code> Requirement 7: <code>/req/core/core-collection-parameter-names</code> Requirement 8: <code>/req/core/core-collection-radius-data-query</code> Requirement 9: <code>/req/core/core-locations-query-response-format</code> Requirement 10: <code>/req/core/core-error-handling</code>

REQUIREMENT 1

IDENTIFIER	<code>/req/core/core-openapi</code>
INCLUDED IN	Requirements class 1: <code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core</code>
STATEMENT	An API definition SHALL be described using an OpenAPI 3.1 document.
A	The API definition SHALL be described using an OpenAPI document (version 3.1 or higher).
B	The OpenAPI document SHALL be encoded as one of the OpenAPI JSON/YAML media types.
C	The landing page SHALL link the OpenAPI document with <code>rel=service-desc</code> .
D	The landing page SHALL link API documentation with <code>rel=service-doc</code> and type <code>text/html</code> .

REQUIREMENT 2

IDENTIFIER /req/core/core-collection-identifier

INCLUDED IN Requirements class 1: <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core>

STATEMENT A collection identifier SHALL NOT convey structured metadata and SHOULD use the fixed value list.

A A collection identifier SHALL NOT be used to convey structured metadata.

B A collection identifier SHOULD use one of `insitu-observations`, `climate_data`, `radar_observations`, `weather_warnings`, `weather_forecast`.

C The identifier string MAY include a postfix to the listed values if needed.

REQUIREMENT 3

IDENTIFIER /req/core/core-collection-title

INCLUDED IN Requirements class 1: <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core>

STATEMENT A title SHALL be set for all collections and SHALL NOT exceed 50 characters.

A A title SHALL be set for all collections.

B A title SHALL NOT have more than 50 characters.

C A title SHOULD be written in English.

D A title SHOULD place the most important information first to guard against truncation.

E A title SHOULD describe the collection understandably for non-experts (topic/domain and geographical area).

REQUIREMENT 4

IDENTIFIER /req/core/core-collection-license

INCLUDED IN Requirements class 1: <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core>

STATEMENT A collection SHALL link to a license of its data with `rel=license` and `type=text/html`.

REQUIREMENT 4

A	The links property in a collection SHALL contain a link to a license of its data.
B	The license link SHALL set the rel property to license.
C	The type property SHALL be set to text/html.
D	If the data is open data, the license SHALL be CC BY 4.0 with href https://creativecommons.org/licenses/by/4.0/ .

REQUIREMENT 5

IDENTIFIER	/req/core/core-collection-spatial-extent
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core
STATEMENT	The collection spatial extent and CRS list SHALL use OGC:CRS84.
A	The value of extent.spatial.crs SHALL be OGC:CRS84.
B	The value of crs SHALL include the element OGC:CRS84.
C	If crs_details is specified for a data_queries object, it SHALL include an object whose crs value is OGC:CRS84.
D	Additional CRSs SHOULD be presented as EPSG:<code> in crs and data_queries.*.crs_details.crs.

REQUIREMENT 6

IDENTIFIER	/req/core/core-collection-temporal-extent
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core
STATEMENT	The value of extent.temporal.trs SHALL be Gregorian.
A	The value of extent.temporal.trs SHALL be Gregorian.

REQUIREMENT 7

IDENTIFIER /req/core/core-collection-parameter-names

INCLUDED IN Requirements class 1: <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core>

STATEMENT Parameter objects SHALL specify label, description and unit using the profile conventions.

A The keys in parameter_names and id in the parameter object SHALL NOT convey structured metadata.

B A parameter object SHALL specify the fields label, description and unit.

C label SHALL NOT have more than 50 characters.

D label SHALL be written in English.

E unit SHALL use QUDT; unit.symbol.type SHALL be <https://qudt.org/vocab/unit/<unit>> and unit.symbol.value the qudt symbol.

F observedProperty SHALL use the CF-convention (observedProperty.id = https://vocab.nerc.ac.uk/standard_name/<name>) when available; otherwise use observedProperty.description.

REQUIREMENT 8

IDENTIFIER /req/core/core-collection-radius-data-query

INCLUDED IN Requirements class 1: <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core>

STATEMENT A radius data query within_units array SHALL contain m (SI unit of length).

A The within_units array in a variables object for a radius data query SHALL contain m.

B within_units MAY contain more values, each defined as a unit at <https://qudt.org> using the unit symbol.

C A radius query with a limit parameter SHOULD return the closest items to the radius centre.

REQUIREMENT 9

IDENTIFIER /req/core/core-locations-query-response-format

INCLUDED IN Requirements class 1: <http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core>

REQUIREMENT 9

STATEMENT	A /locations response SHALL be a GeoJSON FeatureCollection with id and properties.name per feature.
A	The /locations response SHALL be a GeoJSON document with a FeatureCollection object.
B	Each Feature SHALL have id of type string and properties.name of type string.
C	If not all collection parameters are provided, properties.parameter-name SHALL list a subset of the parameter_names keys.
D	Links with more information SHALL be included in properties.links as OGC API – Common link objects.
E	A more detailed description MAY be provided via properties.description.
F	All strings SHALL be in English.

REQUIREMENT 10

IDENTIFIER	/req/core/core-error-handling
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core
STATEMENT	Empty results SHALL return HTTP 204 and 4xx responses SHOULD comply with RFC 9457.
A	A query for data inside the extent that results in no data SHALL return HTTP 204 with no content.
B	A query partially within the extent SHALL respond as within the extent, returning only the in-extent portion.
C	Any 4xx response SHOULD comply with RFC 9457 Problem Details for HTTP APIs.

REQUIREMENTS CLASS 2

IDENTIFIER	http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/insitu-observations
CONFORMANCE CLASS	Conformance class A.2: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/insitu-observations
TARGET-TYPE	MetOcean Profile for OGC API – EDR Profile Standard
NORMATIVE STATEMENTS	Requirement 11: /req/insitu-observations/insitu-collection-data-queries

REQUIREMENTS CLASS 2

Requirement 12: /req/insitu-observations/insitu-collection-parameter-names

Requirement 13: /req/insitu-observations/insitu-collection-custom-dimensions

Requirement 14: /req/insitu-observations/insitu-coveragejson-parameters

REQUIREMENT 11

IDENTIFIER /req/insitu-observations/insitu-collection-data-queries

INCLUDED IN Requirements class 2: <http://www.opengis.net/spec/profile/metocean-ogcap/environmental-data-retrieval/1.0/req/insitu-observations>

STATEMENT An insitu-observations collection SHALL support locations, area and radius.

A A collection SHALL support locations, area and radius.

REQUIREMENT 12

IDENTIFIER /req/insitu-observations/insitu-collection-parameter-names

INCLUDED IN Requirements class 2: <http://www.opengis.net/spec/profile/metocean-ogcap/environmental-data-retrieval/1.0/req/insitu-observations>

STATEMENT Insitu parameters SHALL specify metocean:standard_name, metocean:level and measurementType.

A Each parameter SHALL specify metocean:standard_name set to a value from the standard_name custom dimension.

B Each parameter SHALL specify metocean:level set to a value from the level custom dimension.

C Each parameter SHALL specify measurementType.

REQUIREMENT 13

IDENTIFIER /req/insitu-observations/insitu-collection-custom-dimensions

INCLUDED IN Requirements class 2: <http://www.opengis.net/spec/profile/metocean-ogcap/environmental-data-retrieval/1.0/req/insitu-observations>

STATEMENT An insitu-observations collection SHALL define standard_name and level custom dimensions.

REQUIREMENT 13

A	The collection SHALL have an extent.custom object with id standard_name.
B	Custom dimension standard_name SHALL list CF standard names in values.
C	Custom dimension standard_name SHALL specify reference https://vocab.nerc.ac.uk/standard_name/ .
D	The collection SHALL have an extent.custom object with id level.
E	Custom dimension level SHALL specify reference "Height of measurement above ground level in meters".

REQUIREMENT 14

IDENTIFIER	/req/insitu-observations/insitu-coveragejson-parameters
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/insitu-observations
STATEMENT	CoverageJSON parameters SHALL carry metocean:measurementType, metocean:standard_name and metocean:level.
A	Each object in parameters in a coverage SHALL have metocean:measurementType with the same properties as measurementType.
B	Each object in parameters SHALL have metocean:standard_name.
C	Each object in parameters SHALL have metocean:level.

REQUIREMENTS CLASS 3

IDENTIFIER	http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/nwp
CONFORMANCE CLASS	Conformance class A.3: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/nwp
TARGET-TYPE	MetOcean Profile for OGC API — EDR Profile Standard
NORMATIVE STATEMENTS	Requirement 15: /req/nwp/nwp-collection-parameter-names Requirement 16: /req/nwp/nwp-collection-granularity Requirement 17: /req/nwp/nwp-collection-data-queries

REQUIREMENT 15

IDENTIFIER	/req/nwp/nwp-collection-parameter-names
INCLUDED IN	Requirements class 3: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/nwp
STATEMENT	NWP parameters SHALL use ECMWF short names per Annex D for id and description.
A	WIPPS Manual Global Deterministic NWP parameters SHALL be provided using ECMWF short names per Annex D.
B	The parameter name SHALL be provided in the description property.
C	The ECMWF short name SHALL be provided in the id property.

REQUIREMENT 16

IDENTIFIER	/req/nwp/nwp-collection-granularity
INCLUDED IN	Requirements class 3: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/nwp
STATEMENT	An NWP collection represents a model; instances represent model runs identified by RFC3339 timestamps.
A	A collection SHALL represent and describe a model.
B	A collection instance SHALL represent and describe a model run.
C	A collection instance identifier SHALL be represented by an RFC3339 timestamp.
D	A collection temporal instance or interval SHALL represent a timestep or timesteps of a model run.

REQUIREMENT 17

IDENTIFIER	/req/nwp/nwp-collection-data-queries
INCLUDED IN	Requirements class 3: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/nwp
STATEMENT	An NWP collection SHALL support position and cube; no instance means latest.
A	A collection SHALL support position and cube queries.

REQUIREMENT 17

B	Data queries without an instance SHALL be interpreted as the latest instance of a collection.
C	A datetime parameter between two timestamps SHALL trigger an HTTP 400 Bad Request.

REQUIREMENTS CLASS 4

IDENTIFIER	http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/data-query-response-format
CONFORMANCE CLASS	Conformance class A.4: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/data-query-response-format
TARGET-TYPE	MetOcean Profile for OGC API – EDR Profile Standard
NORMATIVE STATEMENTS	Requirement 18: /req/data-query-response-format/dqrf-encoding Requirement 19: /req/data-query-response-format/dqrf-referencing

REQUIREMENT 18

IDENTIFIER	/req/data-query-response-format/dqrf-encoding
INCLUDED IN	Requirements class 4: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/data-query-response-format
STATEMENT	All data queries SHALL support CoverageJSON as a response format.
A	All data queries SHALL support CoverageJSON as a response format.

REQUIREMENT 19

IDENTIFIER	/req/data-query-response-format/dqrf-referencing
INCLUDED IN	Requirements class 4: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/data-query-response-format
STATEMENT	CoverageJSON domain.referencing system id SHALL reflect the crs query parameter (default OGC: CRS84).
A	With no crs query param, domain.referencing.* system SHALL have type GeographicCRS and id OGC:CRS84.

REQUIREMENT 19

B

With a crs query param, the system object with type GeographicCRS SHALL set id to the crs query param value.



ANNEX A (NORMATIVE) ABSTRACT TEST SUITE



ANNEX A

(NORMATIVE)

ABSTRACT TEST SUITE

CONFORMANCE CLASS A.1

IDENTIFIER	<code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/core</code>
REQUIREMENTS CLASS	Requirements class 1: <code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/core</code>
CONFORMANCE TESTS	<code>Abstract test A.1: /conf/core/core-openapi</code> <code>Abstract test A.2: /conf/core/core-collection-identifier</code> <code>Abstract test A.3: /conf/core/core-collection-title</code> <code>Abstract test A.4: /conf/core/core-collection-license</code> <code>Abstract test A.5: /conf/core/core-collection-spatial-extent</code> <code>Abstract test A.6: /conf/core/core-collection-temporal-extent</code> <code>Abstract test A.7: /conf/core/core-collection-parameter-names</code> <code>Abstract test A.8: /conf/core/core-collection-radius-data-query</code> <code>Abstract test A.9: /conf/core/core-locations-query-response-format</code> <code>Abstract test A.10: /conf/core/core-error-handling</code>

ABSTRACT TEST A.1

IDENTIFIER	<code>/conf/core/core-openapi</code>
REQUIREMENT	Requirement 1: <code>/req/core/core-openapi</code>
TEST PURPOSE	Validate that core openapi is correctly implemented.
TEST METHOD	
STEP	<code>Retrieve the API landing page.</code> <code>Verify a link with rel=service-desc references an OpenAPI 3.1 document.</code> <code>Verify a link with rel=service-doc has type text/html.</code>

ABSTRACT TEST A.2

IDENTIFIER	/conf/core/core-collection-identifier
REQUIREMENT	Requirement 2: /req/core/core-collection-identifier
TEST PURPOSE	Validate that core collection identifier is correctly implemented.
TEST METHOD	
STEP	Retrieve /collections. Verify each collection id matches the fixed value list (optionally with a postfix).

ABSTRACT TEST A.3

IDENTIFIER	/conf/core/core-collection-title
REQUIREMENT	Requirement 3: /req/core/core-collection-title
TEST PURPOSE	Validate that core collection title is correctly implemented.
TEST METHOD	
STEP	Retrieve /collections. Verify each collection has a title of at most 50 characters.

ABSTRACT TEST A.4

IDENTIFIER	/conf/core/core-collection-license
REQUIREMENT	Requirement 4: /req/core/core-collection-license
TEST PURPOSE	Validate that core collection license is correctly implemented.
TEST METHOD	
STEP	Retrieve a collection. Verify links contains a rel=license link with type text/html.

ABSTRACT TEST A.5

IDENTIFIER	/conf/core/core-collection-spatial-extent
REQUIREMENT	Requirement 5: /req/core/core-collection-spatial-extent
TEST PURPOSE	Validate that core collection spatial extent is correctly implemented.
TEST METHOD	
STEP	Retrieve a collection. Verify extent.spatial.crs is OGC:CRS84 and crs includes OGC:CRS84.

ABSTRACT TEST A.6

IDENTIFIER	/conf/core/core-collection-temporal-extent
REQUIREMENT	Requirement 6: /req/core/core-collection-temporal-extent
TEST PURPOSE	Validate that core collection temporal extent is correctly implemented.
TEST METHOD	
STEP	Retrieve a collection. Verify extent.temporal.trs equals Gregorian.

ABSTRACT TEST A.7

IDENTIFIER	/conf/core/core-collection-parameter-names
REQUIREMENT	Requirement 7: /req/core/core-collection-parameter-names
TEST PURPOSE	Validate that core collection parameter names is correctly implemented.
TEST METHOD	
STEP	Retrieve a collection. For each parameter verify label, description and unit are present. Verify unit.symbol.type is a QUDT unit URI and observedProperty.id is a CF standard name URI when present.

ABSTRACT TEST A.8

IDENTIFIER	/conf/core/core-collection-radius-data-query
REQUIREMENT	Requirement 8: /req/core/core-collection-radius-data-query
TEST PURPOSE	Validate that core collection radius data query is correctly implemented.
TEST METHOD	
STEP	Retrieve a collection that supports radius. Verify the radius variables within_units array contains m.

ABSTRACT TEST A.9

IDENTIFIER	/conf/core/core-locations-query-response-format
REQUIREMENT	Requirement 9: /req/core/core-locations-query-response-format
TEST PURPOSE	Validate that core locations query response format is correctly implemented.
TEST METHOD	
STEP	Send a GET /locations query. Verify the response is a GeoJSON FeatureCollection. Verify each Feature has a string id and a string properties.name.

ABSTRACT TEST A.10

IDENTIFIER	/conf/core/core-error-handling
REQUIREMENT	Requirement 10: /req/core/core-error-handling
TEST PURPOSE	Validate that core error handling is correctly implemented.
TEST METHOD	
STEP	Send a query inside the extent that returns no data; verify HTTP 204 with no content.

ABSTRACT TEST A.10

Send a request that yields a 4xx; verify the body complies with RFC 9457 (application/problem+json).

CONFORMANCE CLASS A.2

IDENTIFIER	<code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/insitu-observations</code>
REQUIREMENTS CLASS	Requirements class 2: <code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/insitu-observations</code>
CONFORMANCE TESTS	Abstract test A.11: <code>/conf/insitu-observations/insitu-collection-data-queries</code> Abstract test A.12: <code>/conf/insitu-observations/insitu-collection-parameter-names</code> Abstract test A.13: <code>/conf/insitu-observations/insitu-collection-custom-dimensions</code> Abstract test A.14: <code>/conf/insitu-observations/insitu-coveragejson-parameters</code>

ABSTRACT TEST A.11

IDENTIFIER	<code>/conf/insitu-observations/insitu-collection-data-queries</code>
REQUIREMENT	Requirement 11: <code>/req/insitu-observations/insitu-collection-data-queries</code>
TEST PURPOSE	Validate that insitu collection data queries is correctly implemented.
TEST METHOD	
STEP	Retrieve the insitu-observations collection. Verify data_queries contains locations, area and radius.

ABSTRACT TEST A.12

IDENTIFIER	<code>/conf/insitu-observations/insitu-collection-parameter-names</code>
REQUIREMENT	Requirement 12: <code>/req/insitu-observations/insitu-collection-parameter-names</code>
TEST PURPOSE	Validate that insitu collection parameter names is correctly implemented.
TEST METHOD	

ABSTRACT TEST A.12

STEP	Retrieve the insitu-observations collection.
	Verify each parameter has metocean:standard_name, metocean:level and measurementType.

ABSTRACT TEST A.13

IDENTIFIER	/conf/insitu-observations/insitu-collection-custom-dimensions
REQUIREMENT	Requirement 13: /req/insitu-observations/insitu-collection-custom-dimensions
TEST PURPOSE	Validate that insitu collection custom dimensions is correctly implemented.
TEST METHOD	
STEP	Retrieve the insitu-observations collection.
	Verify extent.custom includes standard_name (with the nerc reference) and level.

ABSTRACT TEST A.14

IDENTIFIER	/conf/insitu-observations/insitu-coveragejson-parameters
REQUIREMENT	Requirement 14: /req/insitu-observations/insitu-coveragejson-parameters
TEST PURPOSE	Validate that insitu coveragejson parameters is correctly implemented.
TEST METHOD	
STEP	Request insitu data as CoverageJSON.
	Verify each parameters object has metocean:measurementType, metocean:standard_name and metocean:level.

CONFORMANCE CLASS A.3

IDENTIFIER	http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/nwp
REQUIREMENTS CLASS	Requirements class 3: http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/nwp

CONFORMANCE CLASS A.3

CONFORMANCE TESTS

Abstract test A.15: `/conf/nwp/nwp-collection-parameter-names`
Abstract test A.16: `/conf/nwp/nwp-collection-granularity`
Abstract test A.17: `/conf/nwp/nwp-collection-data-queries`

ABSTRACT TEST A.15

IDENTIFIER `/conf/nwp/nwp-collection-parameter-names`

REQUIREMENT Requirement 15: `/req/nwp/nwp-collection-parameter-names`

TEST PURPOSE Validate that nwp collection parameter names is correctly implemented.

TEST METHOD

STEP Retrieve the NWP collection.
Verify parameter ids use ECMWF short names from Annex D and descriptions carry the parameter name.

ABSTRACT TEST A.16

IDENTIFIER `/conf/nwp/nwp-collection-granularity`

REQUIREMENT Requirement 16: `/req/nwp/nwp-collection-granularity`

TEST PURPOSE Validate that nwp collection granularity is correctly implemented.

TEST METHOD

STEP Retrieve `/collections/weather_forecast/instances`.
Verify each instance id is an RFC3339 timestamp representing a model run.

ABSTRACT TEST A.17

IDENTIFIER `/conf/nwp/nwp-collection-data-queries`

REQUIREMENT Requirement 17: `/req/nwp/nwp-collection-data-queries`

TEST PURPOSE Validate that nwp collection data queries is correctly implemented.

ABSTRACT TEST A.17

TEST METHOD

	Retrieve the NWP collection.
STEP	Verify data_queries contains position and cube.
	Verify a datetime range request triggers HTTP 400.

CONFORMANCE CLASS A.4

IDENTIFIER	<code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/conf/data-query-response-format</code>
REQUIREMENTS CLASS	Requirements class 4: <code>http://www.opengis.net/spec/profile/metocean-ogcapi-environmental-data-retrieval/1.0/req/data-query-response-format</code>
CONFORMANCE TESTS	Abstract test A.18: <code>/conf/data-query-response-format/dqrf-encoding</code> Abstract test A.19: <code>/conf/data-query-response-format/dqrf-referencing</code>

ABSTRACT TEST A.18

IDENTIFIER	<code>/conf/data-query-response-format/dqrf-encoding</code>
REQUIREMENT	Requirement 18: <code>/req/data-query-response-format/dqrf-encoding</code>
TEST PURPOSE	Validate that dqrf encoding is correctly implemented.
TEST METHOD	
STEP	For each data query, verify CoverageJSON is an offered response format.

ABSTRACT TEST A.19

IDENTIFIER	<code>/conf/data-query-response-format/dqrf-referencing</code>
REQUIREMENT	Requirement 19: <code>/req/data-query-response-format/dqrf-referencing</code>
TEST PURPOSE	Validate that dqrf referencing is correctly implemented.
TEST METHOD	

ABSTRACT TEST A.19

STEP

Request data without a crs param; verify domain.referencing system id is OGC:CRS84.

Request data with a crs param; verify the system id matches the crs param.